
EXERCISE

Lecture 3: Optimization Recap

AI508 — Algorithmic Game Theory

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1 Convex Sets

Any line-segment with fixed ends inside the set, should only be in the set.

2 Convex Functions

Any line-segment with fixed ends on the function, should be above the function itself.

$$\nabla^2 f \succeq 0$$

2.1 Strongly Convex

$$\nabla^2 f \succeq \alpha I$$

3 Probability Simplex

$$\Delta^{n-1} = \left\{ (x_0, \dots, x_{n-1}) \in \mathbb{R}^n \mid \sum_{j=0}^{n-1} x_j = 1, x_j \geq 0, \forall j \in 0, 1, \dots, n-1 \right\}$$

This is helpful for probabilities given that for probabilities its mostly true that $\sum_{p \in P} p_i = 1$

4 Gradient Descent

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k)$$

$$x^* \arg \min_x f(x)$$

or

$$x^* \arg \min_{k=0,1,\dots,T-1} f(x_k)$$

4.1 Projected Gradient Descent

$$y_{k+1} = x_k - \alpha_k \nabla f(x_k)$$

$$x_{k+1} = \Pi_{\mathcal{X}}(y_k)$$

where $\Pi_{\mathcal{X}}$ is the euclidean projection back into the feasible set.

5 LP Formulation

$$\begin{aligned} \max c^T x \\ Ax \leq B \end{aligned}$$

Where the constants are always linear.